

Draft ACM Master's Thesis Proposal for 625.803-804

Student: Aditya Bhatt

JHU Email: abhatt25@jh.edu

Phone: 862-251-2956

Program: M.S. in Applied and Computational Mathematics

Course Sequence: 625.803-804 Applied and Computational Mathematics Master's Thesis

Semester/Term: Summer 2026 - Fall 2026

Research Adviser: Dr. Zerotti Woods, zerotti.woods@jhuapl.edu

Proposed Second Reader: Professor Moustapha Pemy, mpemy@towson.edu

Academic Adviser: Professor Beryl Castello, beryl.castello@jhu.edu

Working Title: Overthinking Boundary in Reasoning Large Language Models

1. Problem Motivation

Reasoning-oriented large language models often improve when they are allowed to generate intermediate steps before committing to a final answer. Empirically, however, the marginal value of additional reasoning is not uniformly positive. Beyond a model- and task-dependent point, continued inference can revise a correct answer, drift away from the relevant solution path, or consume additional compute without improving correctness. This phenomenon is commonly described as overthinking.

The central problem of this thesis is to identify when additional test-time reasoning stops being beneficial and starts becoming harmful. That turning point will be called the overthinking boundary. The project asks whether this boundary can be defined mathematically, detected empirically, and used to support a practical stopping rule for reasoning models.

This topic fits naturally within applied and computational mathematics because it combines stochastic modeling, sequential decision-making, hazard-based analysis, uncertainty quantification, and empirical evaluation of computational systems. If a model can be stopped near the boundary rather than forced to continue reasoning indefinitely, the result may be better accuracy, lower compute cost, and more reliable behavior.

2. Proposed Research Question and Objectives

The main research question is:

Can the point at which further reasoning becomes counterproductive in a reasoning language model be characterized as a mathematically meaningful stopping boundary and estimated from observable trace features?

The thesis will pursue four objectives: (1) formalize a continuation-value framework for reasoning traces using current correctness belief, repair probability, corruption probability, and per-step compute cost; (2) study whether the resulting drift admits a one-crossing structure that defines an operational stopping boundary; (3) estimate the relevant quantities from empirical traces using observables such as answer revisions, entropy-related signals, and hidden-state-derived proxies where available; and (4) compare the resulting stopping rules against fixed-length, never-stop, and alternative sequential-detection baselines.

3. Background and Original Contribution

Recent literature on reasoning models and adaptive test-time compute suggests that longer chain-of-thought is not uniformly beneficial. Work on reasoning-token budgets, adaptive inference, and stopping heuristics indicates that extra computation can help in some regimes but waste or even harm performance in others; however, much of this literature is driven by proxy scores or engineering heuristics rather than an explicit continuation-value formulation that separates the chance of repairing a wrong answer from the chance of corrupting a correct one. This proposal builds on that literature while aiming for a more principled mathematical treatment of the stopping problem.

The intended contribution is threefold. First, the thesis formalizes overthinking as an optimal stopping problem with an explicit one-step continuation value rather than as a loosely described side effect of long reasoning traces. Second, it separates beneficial continuation from harmful continuation through distinct repair and corruption hazards. Third, it evaluates these ideas on real traces from open-weight reasoning models, so the theory is tested against observable model behavior rather than only toy examples. The project therefore goes beyond straightforward implementation of known methods and is intended to expand knowledge in applied mathematics at the intersection of stochastic processes, sequential inference, and modern AI systems.

4. Proposed Methodology

The project will begin with a focused literature survey on four areas: overthinking in reasoning models, adaptive test-time compute, sequential stopping rules, and hazard- or drift-based decision models. That survey will clarify the state of the art and identify observables most plausible for estimating the value of continued reasoning.

The analytical core models each reasoning trace as a discrete-time stochastic process on a filtration generated by the observable history of intermediate outputs and derived trace features. Let A_t denote the model's answer after step t , let $C_t = \mathbf{1}\{A_t = Y^*\}$ denote correctness relative to ground truth, and let $q_t = \mathbb{P}(C_t = 1 \mid \mathcal{F}_t)$ be the current correctness belief. The one-step continuation value will be written as

$$\mu_t = (1 - q_t)\alpha_t - q_t\beta_t - \lambda,$$

where α_t is the repair hazard, β_t is the corruption hazard, and λ is a per-step compute cost. The overthinking boundary is the first step at which $\mu_t \leq 0$. The thesis will study conditions under which this drift has a one-crossing structure and will examine statistical procedures for estimating the boundary from observable features when correctness is not directly observed during inference.

The empirical component will use real traces generated by open-weight reasoning models, currently centered on DeepSeek-R1-Distill-Qwen-1.5B and Qwen2.5-0.5B-Instruct. Benchmark tasks will focus on mathematical question answering, beginning with GSM8K-style problems and related reasoning tasks. For each run, the

project will collect stepwise outputs, answer revisions, entropy-related quantities, and hidden-state-derived proxies where available. The study will compare hazard-based drift rules, sequentially valid detector variants, fixed-length policies, and never-stop baselines using answer accuracy, average stopping step, compute efficiency, and gap to an oracle hindsight stopping point.

5. Expected Results and Significance

The expected outcome is a mathematically rigorous and empirically validated framework showing that overthinking can be analyzed as a stopping phenomenon rather than only as an anecdotal behavior of large language models. The thesis aims to identify conditions under which reasoning traces move from a beneficial regime to a harmful regime and to show that a principled stopping rule can outperform naive always-continue policies.

If successful, the work will contribute both theoretical and practical value. Theoretically, it will connect reasoning-model behavior to stochastic processes, optimal stopping, and sequential statistics. Practically, it will provide a basis for reducing inference cost and avoiding answer degradation in reasoning systems.

6. Final Documentation Plan

This project is being proposed under the thesis option, 625.803-804, rather than the research-paper option. The planned final deliverable is a master's thesis submitted in accordance with the Johns Hopkins electronic thesis and dissertation requirements for the MSE Library, together with the required public defense presentation. The expected final authorship is the student as sole author of the thesis, with guidance from the research adviser and feedback from the second reader. If the work later yields a conference or journal paper, possible coauthorship will be discussed separately, but no additional coauthors are assumed for the thesis proposal itself.

7. Preliminary Work and Semester Plan

Substantial preliminary work has already been completed. The project repository includes an initial mathematical framework for the overthinking boundary, an expanded literature synthesis, pilot and larger-scale empirical results, and supporting analysis code. Early results indicate that at least some model-task pairs exhibit a measurable boundary after which continued reasoning reduces utility, which supports the feasibility of the thesis within the two-semester structure.

During 625.803, the focus will be on refining the literature review, finalizing the mathematical formulation, and extending the empirical pipeline as needed while preparing the required semester status report. During 625.804, the focus will be on final experiments, robustness analysis, thesis writing, revision of the full document, and preparation of the defense presentation.